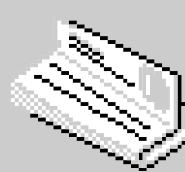
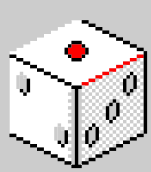
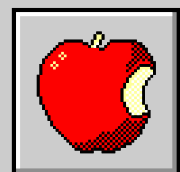


CSE6344

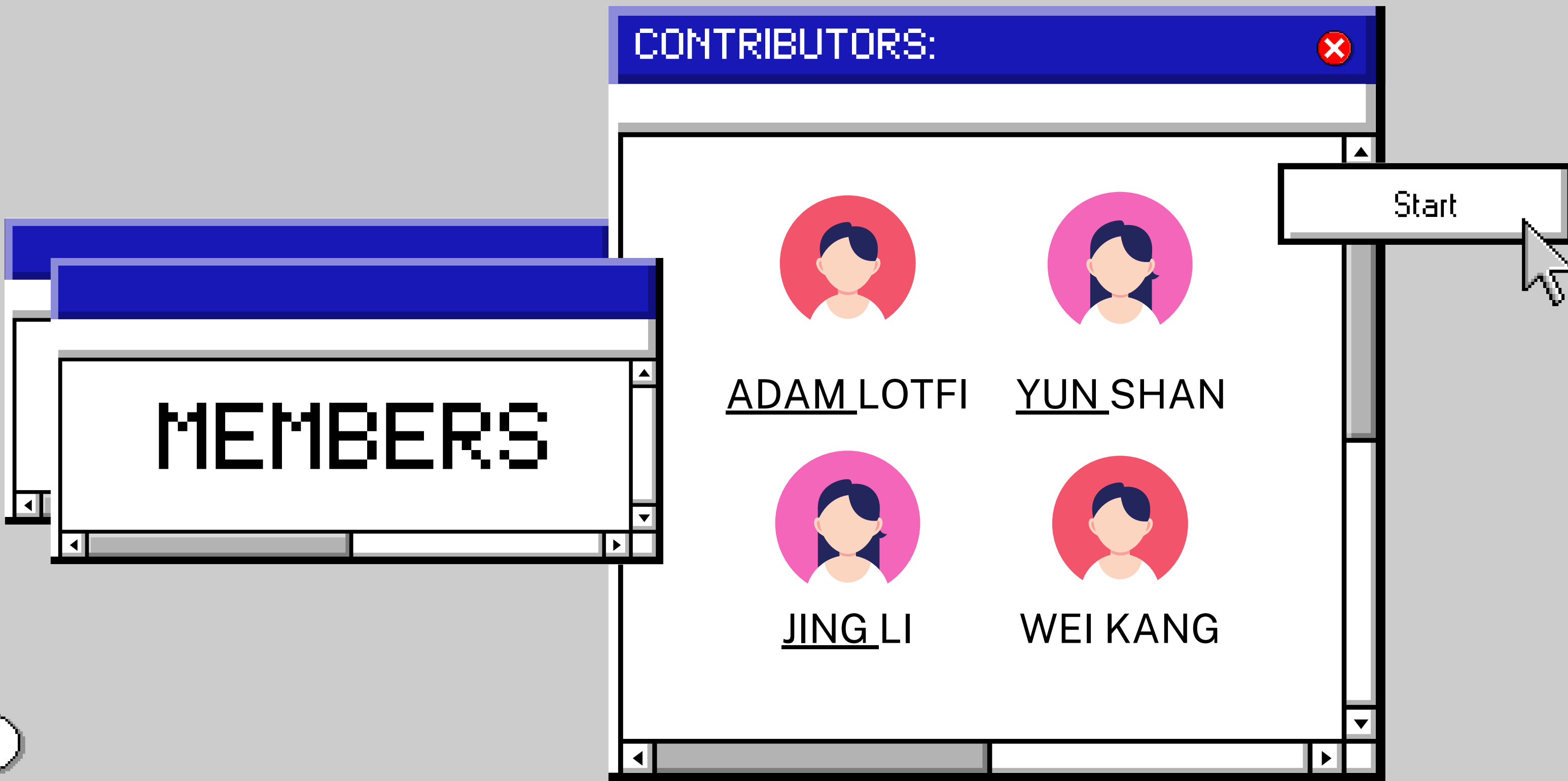
ASSIGNMENT 1



BY GROUP 1



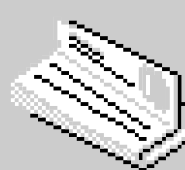
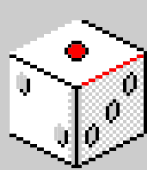
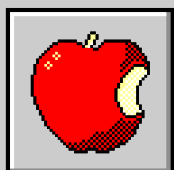
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OVERVIEW

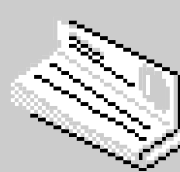
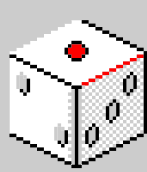
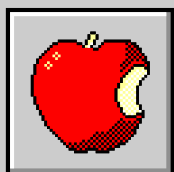


GIVEN STRINGS
STRING DISTRIBUTIONS
REGEX CONSTRUCTION
NFA BUILDING
DFA CONVERSION
COMPILED DFA
TESTS



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GIVEN STRINGS, DISTRIBUTIONS





INDIVIDUAL TASKS



ADAM'S SET 1

GIVEN STRING:

gx2, gx4, gx6, gxg

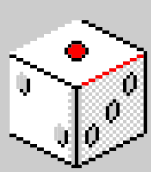
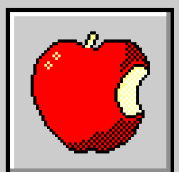
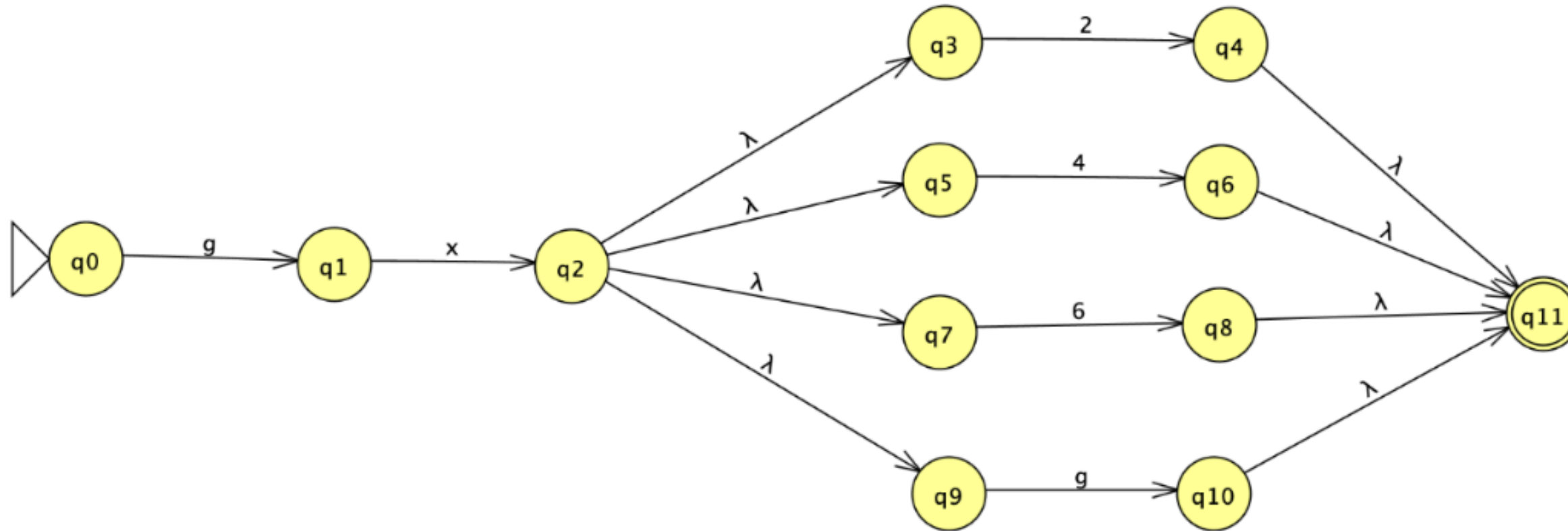
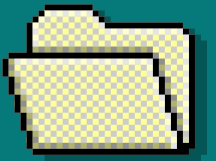


REGULAR
EXPRESSION:

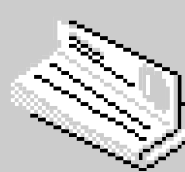
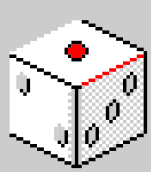
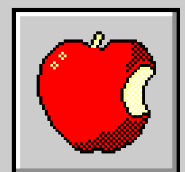
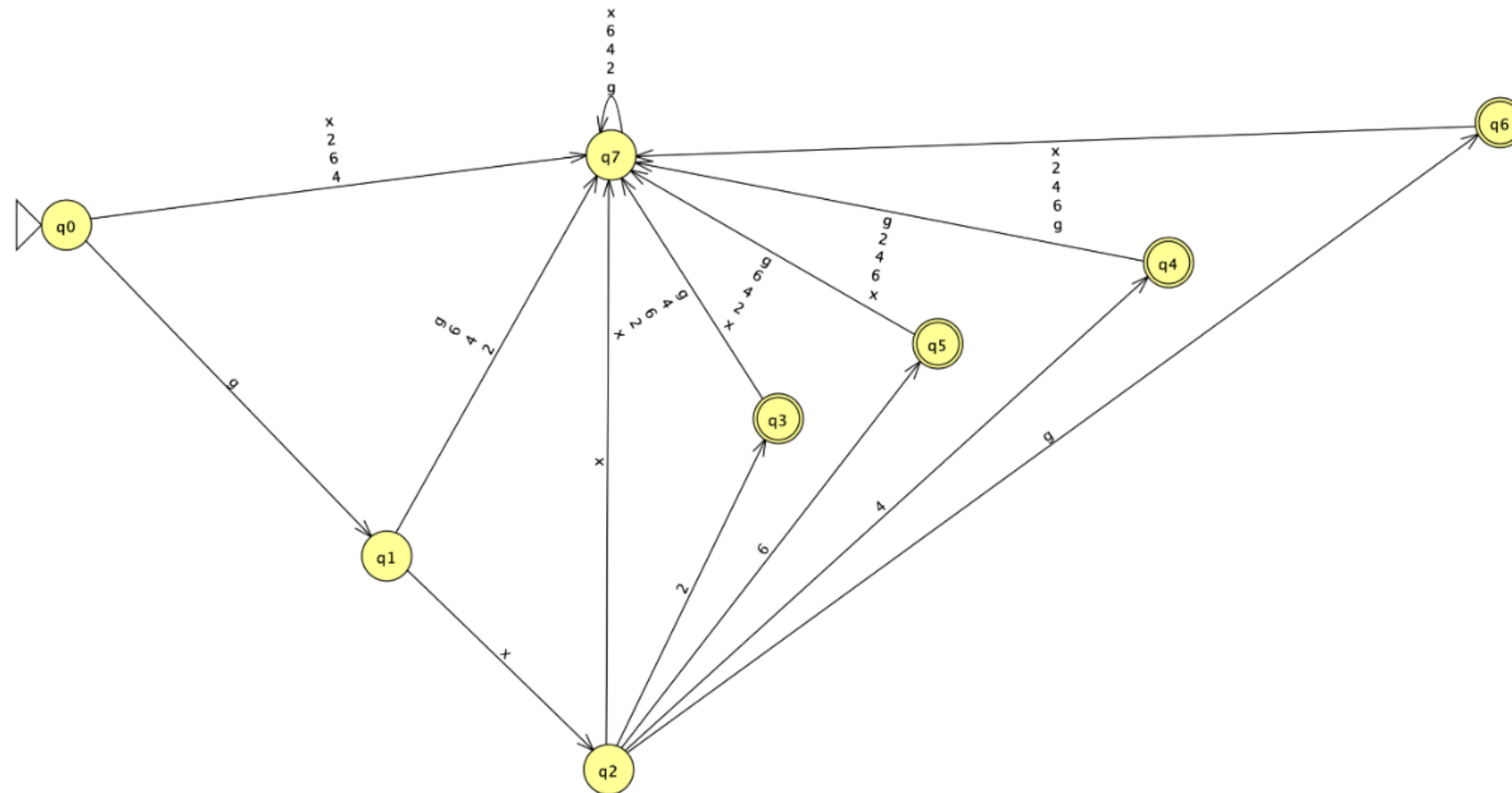
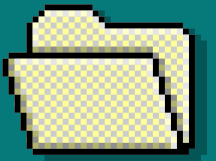
`gx(2U4U6Ug)`



REGEX $\rightarrow$ NFA



NFA $\rightarrow$ DFA



INDIVIDUAL TESTS

Input	Result
gx2	Accept
gx4	Accept
gx6	Accept
gxg	Accept
gxx	Reject
xxg	Reject
2xg	Reject
xxx	Reject
123	Reject
xg*	Reject

(TESTS WERE RAN IN JFLAP 7.1)

WEI KANG'S SET 2

GIVEN STRING:

n2, n4, n6, ng

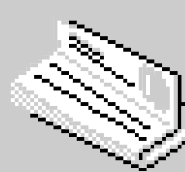
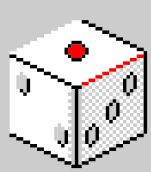
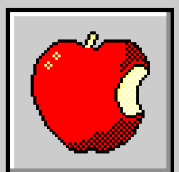
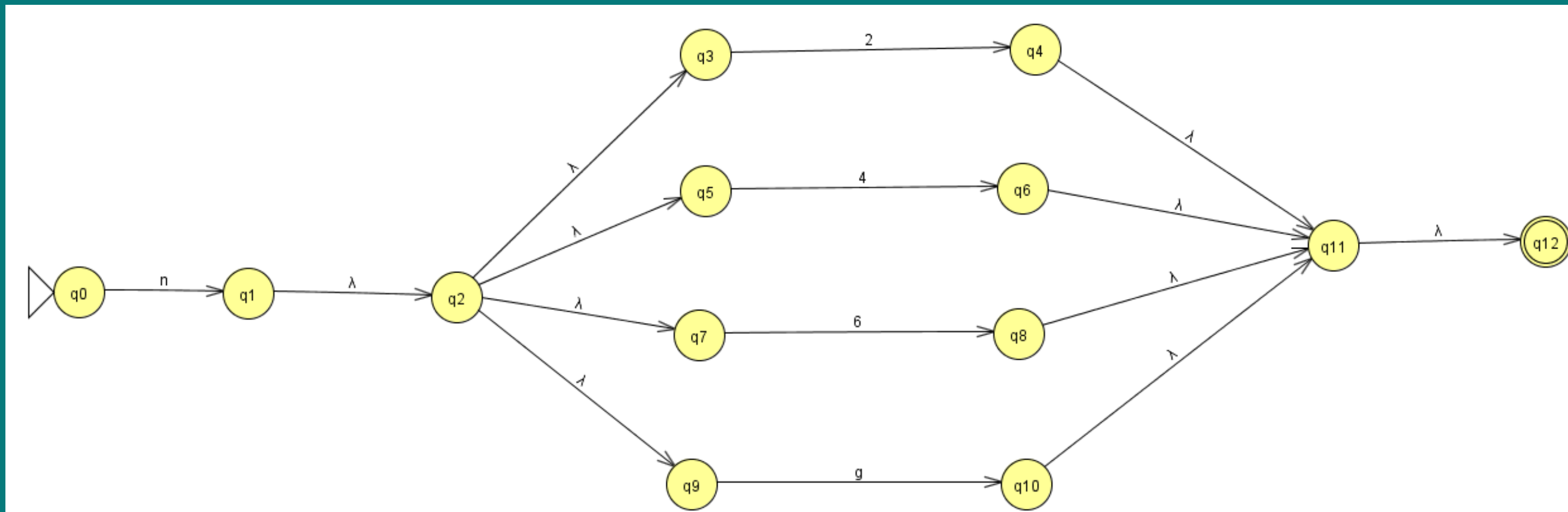
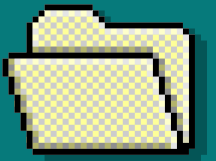


REGULAR EXPRESSION:

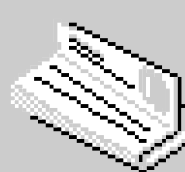
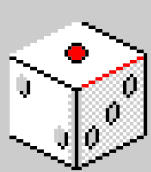
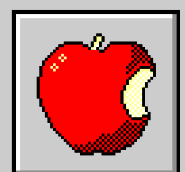
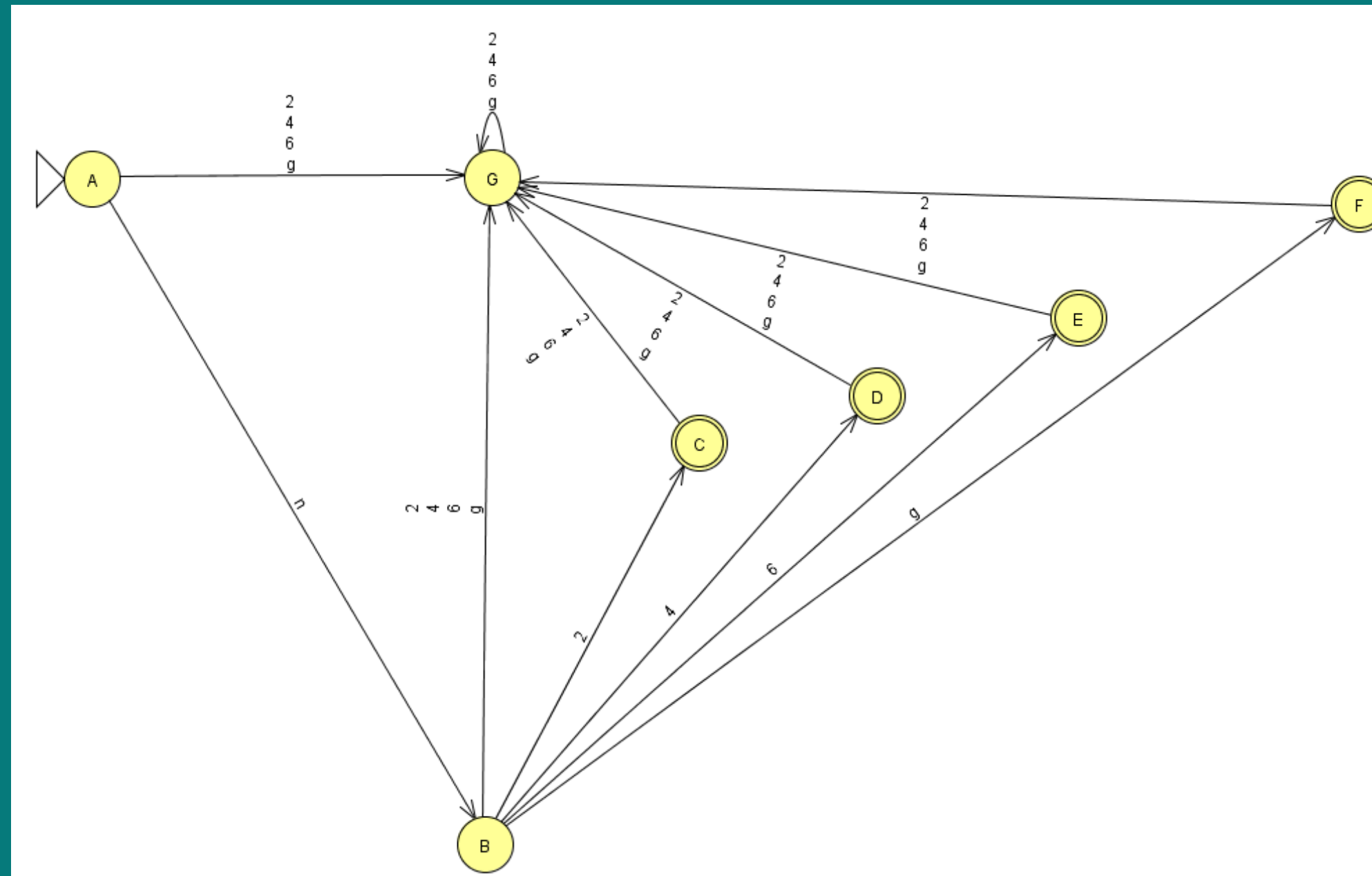
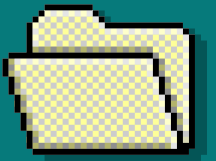
$n(2 \cup 4 \cup 6 \cup g)$



REGEX $\rightarrow$ NFA



NFA → DFA



INDIVIDUAL TESTS

Input	Result
n2	Accept
n4	Accept
n6	Accept
ng	Accept
nn	Reject
n7	Reject
gn	Reject
n24	Reject

(TESTS WERE RAN IN JFLAP 7.1)

YUN SHAN'S SET 3

GIVEN STRING:

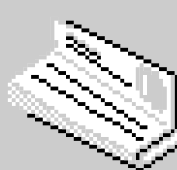
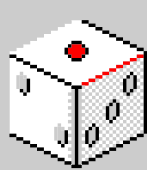
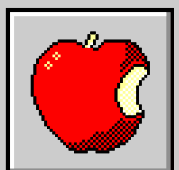
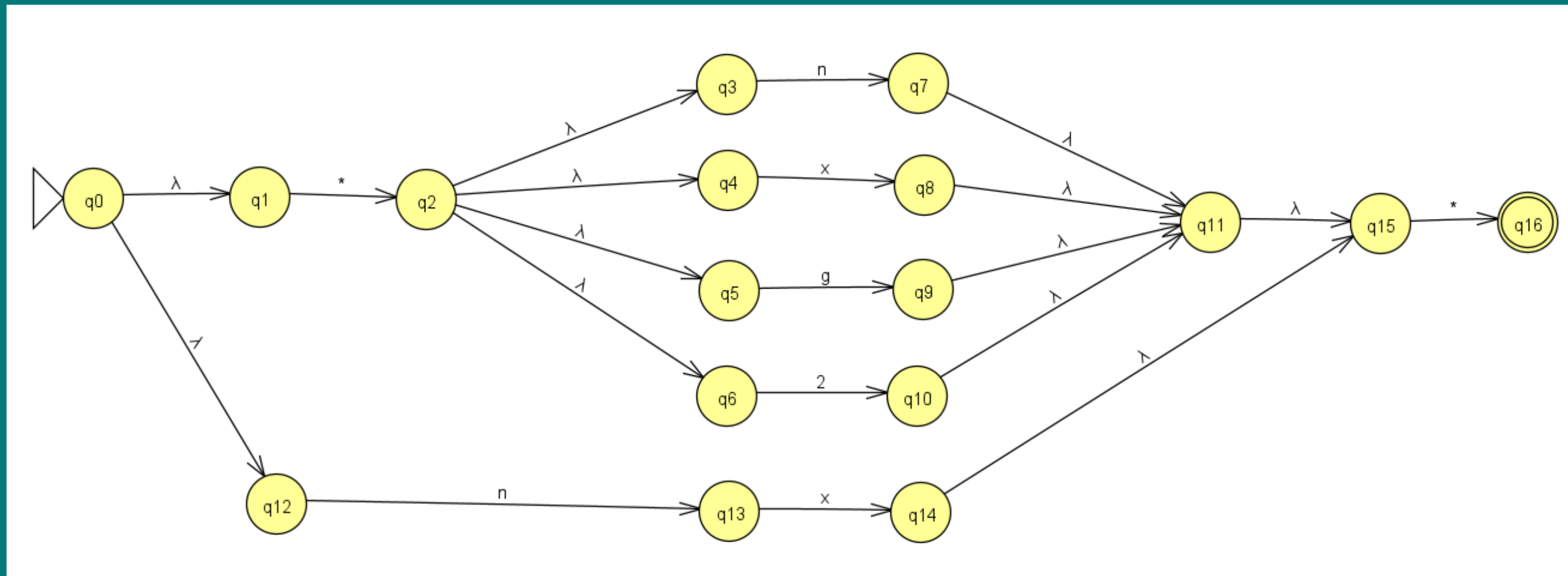
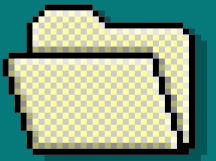
$*n^*, *x^*, *g^*, *2^*, nx^*$



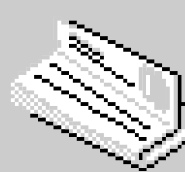
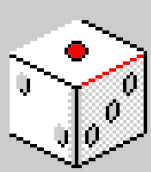
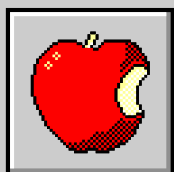
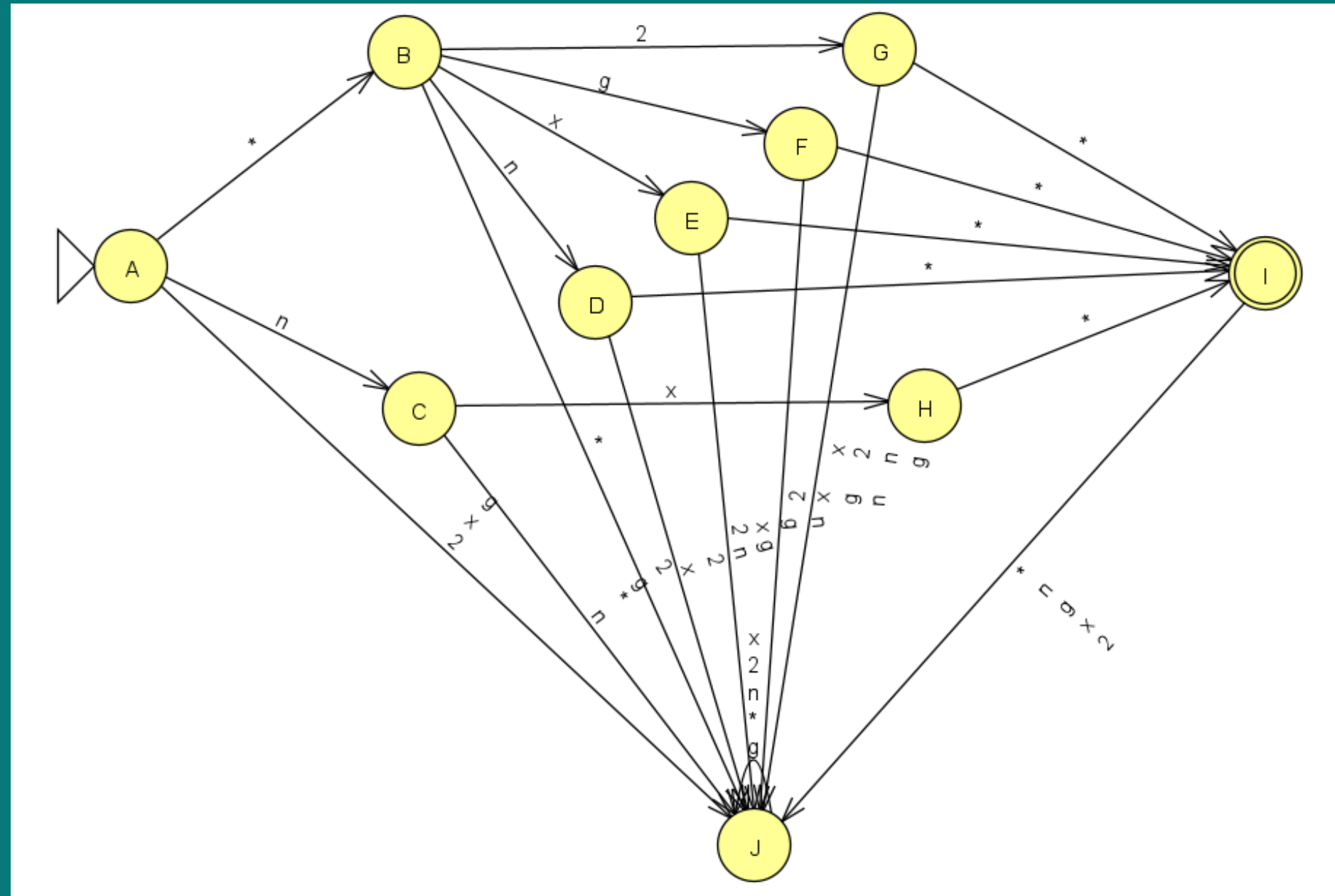
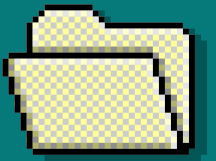
REGULAR EXPRESSION:

$$[* (n \cup x \cup g \cup 2) \cup nx]^*$$


REGEX → NFA



NFA → DFA



INDIVIDUAL TESTS

Input	Result
n^*	Accept
x^*	Accept
g^*	Accept
2^*	Accept
nx^*	Accept
n	Reject
x	Reject
g	Reject
2	Reject
$*$	Reject

JING LI'S SET 4

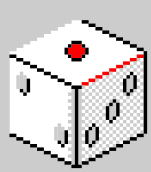
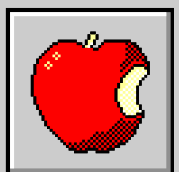
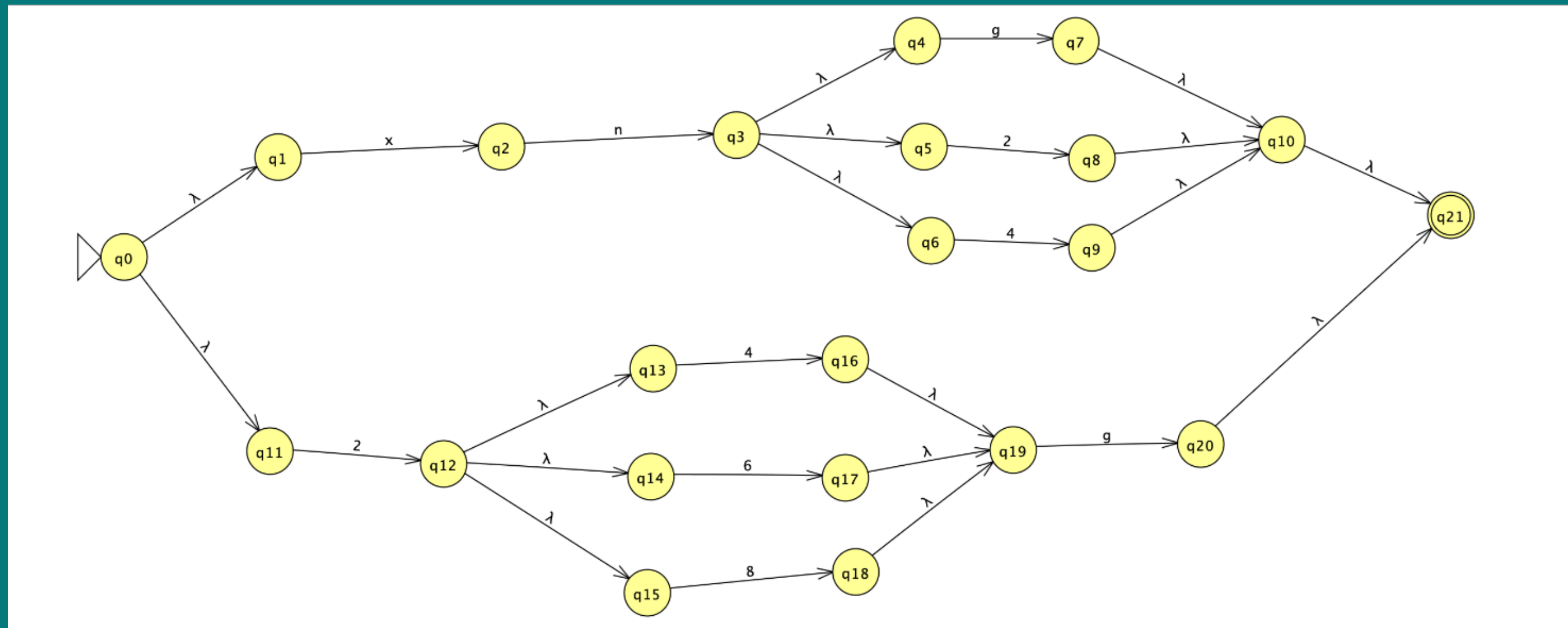
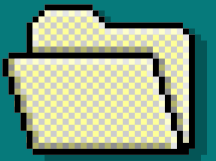


xng, xn2, xn4, 24g, 26g, 28g

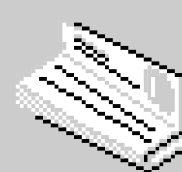
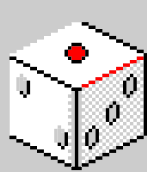
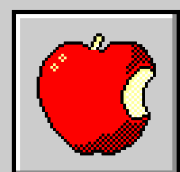
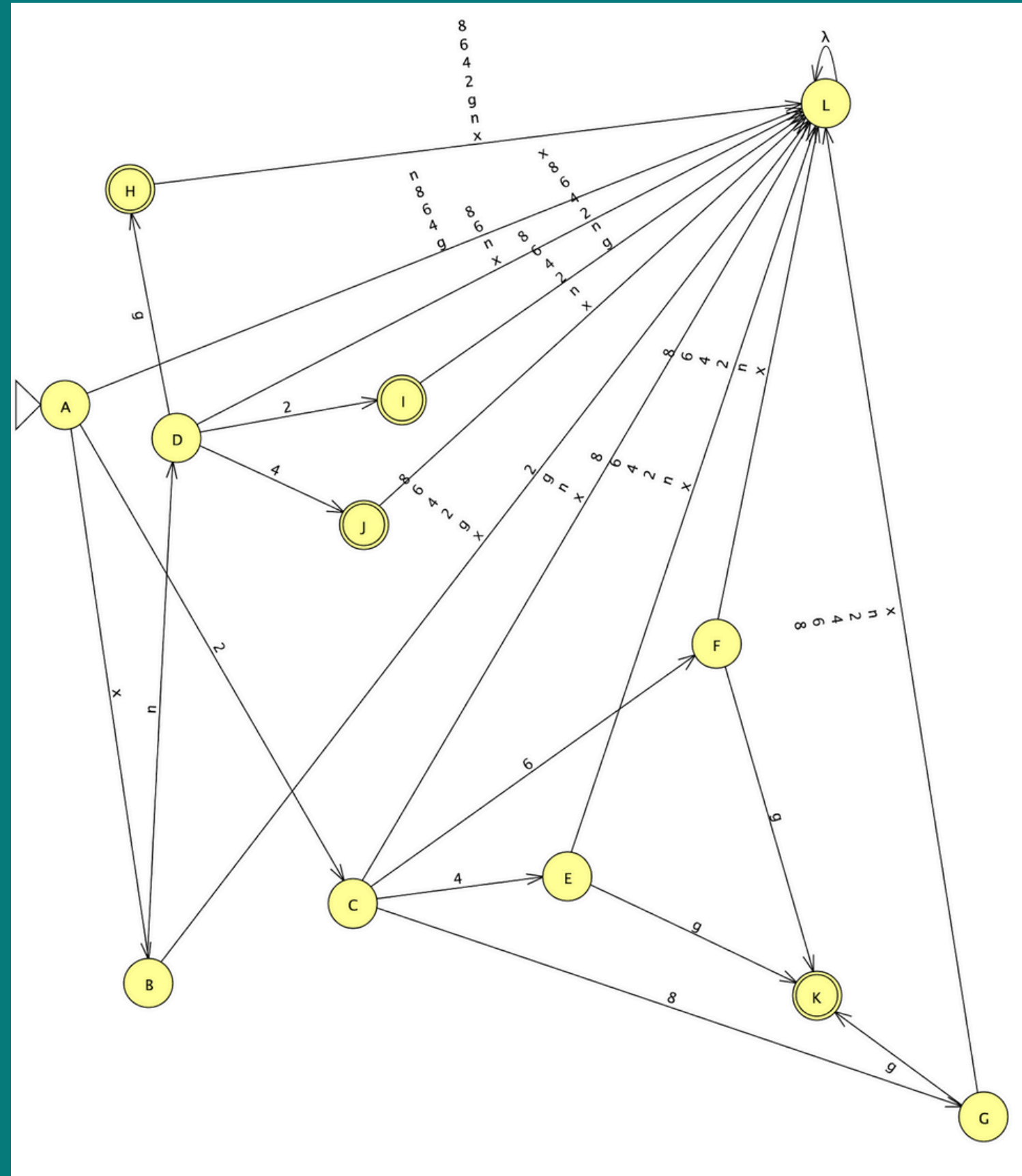
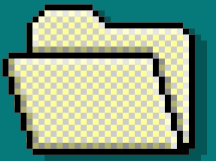
REGULAR EXPRESSION:

$$(x_n(g \cup 2 \cup 4)) \cup (2(4 \cup 6 \cup 8)g)$$


REGEX → NFA



NFA → DFA



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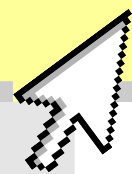
INDIVIDUAL TESTS

Input	Result
xng	Accept
xn2	Accept
xn4	Accept
24g	Accept
26g	Accept
28g	Accept
x	Reject
xn	Reject
2	Reject
24	Reject
26	Reject
xng2	Reject

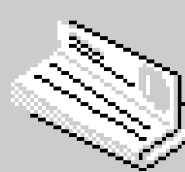
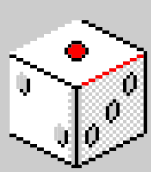
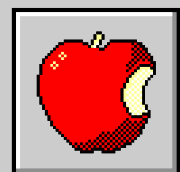
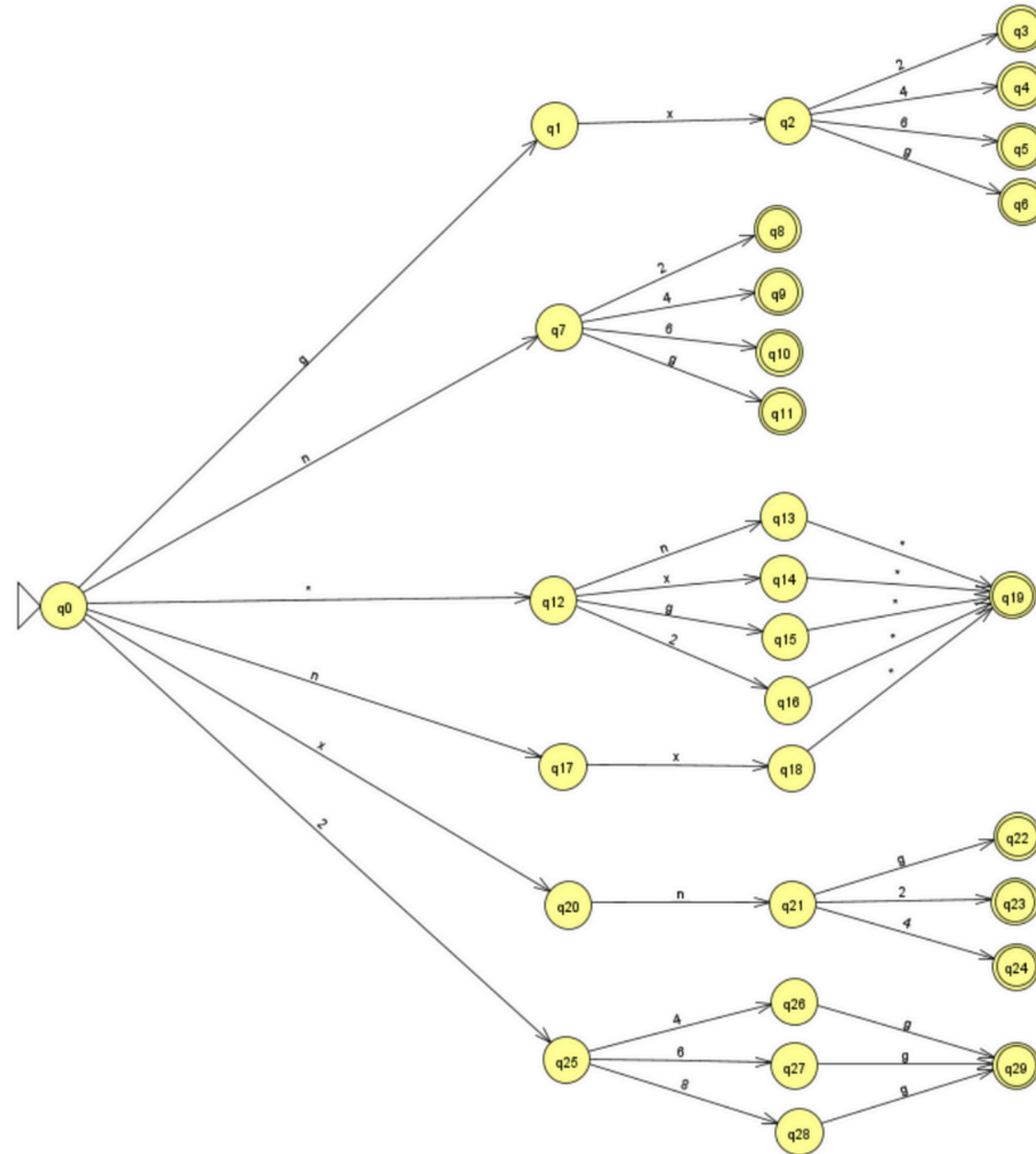
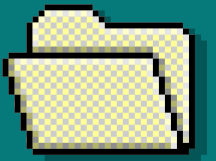
(TESTS WERE RAN IN JFLAP 7.1)



COMBINED DFA



DFA



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TESTS

Input	Result	Input	Result
n	Accept	gxx	Reject
x	Accept	xxg	Reject
g	Accept	2xg	Reject
2	Accept	xxx	Reject
gx2	Accept	xg*	Reject
gx4	Accept	nn	Reject
gx6	Accept	n7	Reject
gxg	Accept	gn	Reject
n2	Accept	n24	Reject
n4	Accept	*n	Reject
n6	Accept	**x	Reject
ng	Accept	g*	Reject
xng	Accept	*2**	Reject
xn2	Accept	*	Reject
xn4	Accept	x	Reject
nx*	Accept	xn	Reject
24g	Accept	2	Reject
26g	Accept	24	Reject
28g	Accept	26	Reject
		xng2	Reject

(TESTS WERE RAN IN JFLAP 7.1)

